Unit 1, Lesson 7: Division of Whole Numbers

Benchmark

MA.5.NSO.2.2 Divide multi-digit whole numbers, up to five digits by two digits, including using a standard algorithm with procedural fluency. Represent remainders as fractions.

Learning Target

- ☐ I can find whole-number quotients of whole numbers.

1.7.1 Warm-Up: What Is Math Anxiety?

Have you ever sat down to take a math test and immediately felt your heart beat faster and your palms start to sweat? Sometimes this can happen if you are feeling anxious about math.

1. What are some of your worries about math class this year?

Maryam Mirzakhani is an award-winning mathematician who was born in Iran. After being the first woman to win the prestigious Fields Medal, she stated in an interview, “I did poorly in math for a couple of years in middle school; I was just not interested in thinking about it. I can see that without being excited mathematics can look pointless and cold. The beauty of mathematics only shows itself to more patient followers.”

2. Remember: the brain is flexible, so your math skills can always grow and develop. Write a positive statement about learning math this year.

1.7.2 Guided Instruction: Division of Whole Numbers

Division is
- ○ repeated addition.
- ○ repeated subtraction.

This is because when dividing, we are **partitioning**, or taking out, equal groups from the whole.

1. A division expression is shown.

$$253 \div 12$$

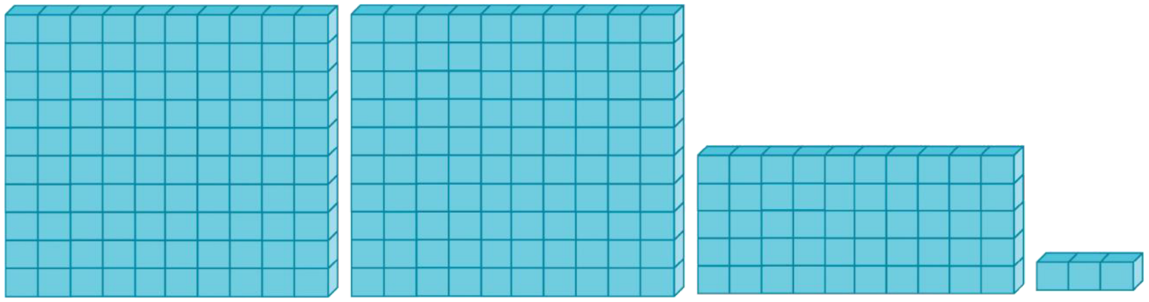
The ***divisor*** is the size of the groups that are being partitioned.

a. In the expression, the divisor is
- ○ 253.
- ○ 12.

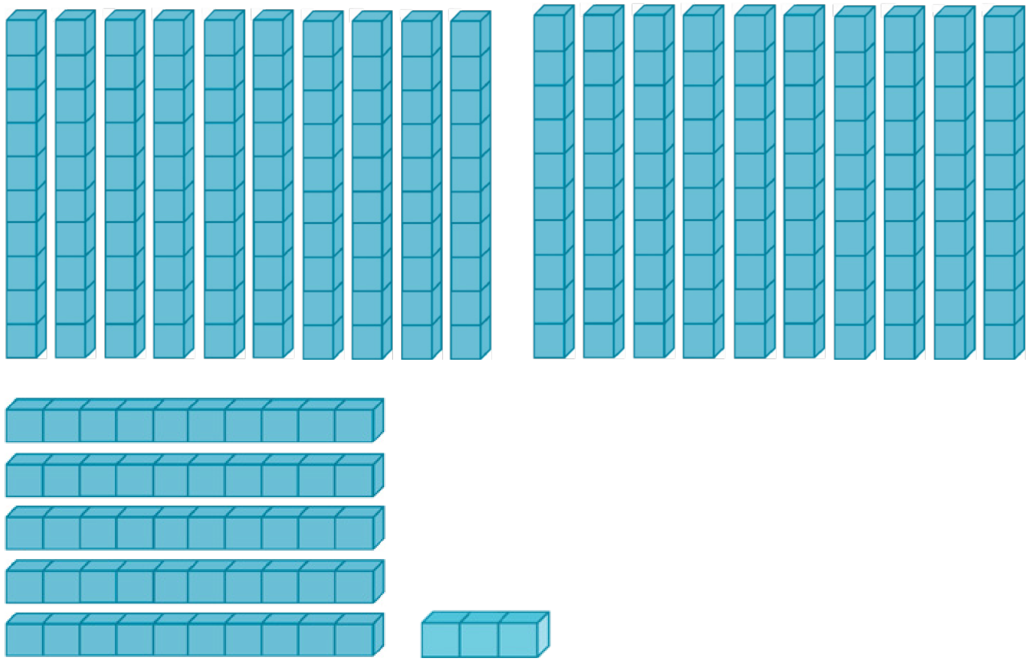
The ***dividend*** is the group that is being partitioned.

b. In the expression, the dividend is
- ○ 253.
- ○ 12.

A base-ten model of 253 is shown.



The model should be arranged in 12 equal groups. Since there are not 12 hundreds flats, regroup the hundreds-flats into ten-rods.



2. Partition the ten-rods into 12 equal groups by sketching the ten-rods in the boxes.

3. How many tens are left after partitioning?

The remaining ten-rods and the 3 ones should be partitioned by regrouping the ten-rods into ones.

4. Add to your sketch in the boxes the partitioning of the ones.

5. How many ones were not able to be partitioned?

The one-blocks that were not able to be partitioned are the **remainder**. The amount left over can also be written as a fraction, $\frac{remaining\ one\ blocks}{divisor}$.

6. Write the remainder as a fraction.

7. What is the value of the base-ten blocks in each group?

When there is a remainder, the **quotient** is the value of the base ten-blocks and the remainder written as a fraction.

8. Write the quotient.

9. Complete the equation.

$253 \div 12 =$

1.7.3 Guided Instruction: Partial Quotient

Find $2{,}743 \div 14$ using the **partial quotient** model. The partial quotient model takes out (subtracts) parts of the total quotient each time. The number of groups that are taken out each time doesn't matter; the quotient will be the same.

1. Complete the table.

Number of Groups of 14	Total Taken Out
1	14
2	28
3	42
4	56
5	70
10	140
20	
30	420
40	
50	
100	1,400
200	
300	4,200
400	
500	7,000

2. What two numbers in the table is $2{,}743$ between?

3. Use the smaller number of groups of 14 to begin the repeated subtraction. Add rows if necessary, but do not use rows if there are no remaining groups of 14.

	Description
$2{,}743$	
$-$ ______	______ groups of 14
	Amount that remains
$-$ ______	______ groups of 14
	Amount that remains
$-$ ______	______ groups of 14
	Amount that remains
$-$ ______	______ groups of 14
	Amount that remains
$-$ ______	______ groups of 14
	Amount that remains

4. How many groups of 14 were subtracted?

5. What amount remains?

6. Write the quotient using the remainder notation and as a mixed number.

1.7.4 Try It: Division of Whole Numbers

1. Complete the table to find $8{,}988 \div 25$ using a modified partial quotient model.

Type of Group	Number of Groups of 25	Total Taken Out
ones	1	25
ones	2	50
ones	3	
ones	4	
ones	5	
	etc.	
tens	10	250
tens	20	
tens	30	750
tens	40	
tens	50	
	etc.	
hundreds	100	$2{,}500$
hundreds	200	
hundreds	300	$7{,}500$
hundreds	400	
hundreds	500	
	etc.	

2. What two hundreds groups in the table is $8{,}988$ between?

3. Use the smaller number of hundreds groups of 25 to begin.

$8,988$	**Description**
	Number of hundreds groups of 25
$-$ ______	______
	Amount that remains
	Number of tens groups of 25
$-$ ______	______
	Amount that remains
	Number of ones groups of 25
$-$ ______	______
	Amount that remains

4. What is the quotient of $8,988 \div 25$? Write your answer using a remainder.

Another way to show this type of partial quotient is with long division.

5. Use the partial quotient model from above to complete the long division. Notice that the quotient is set up to write the remainder as the partial group of 25.

$$\frac{\square}{25}$$

$$25\overline{)8,988}$$

Number of hundreds groups of 25

Amount that remains

Number of tens groups of 25

Amount that remains

Number of ones groups of 25

Amount that remains

1.7.5 Your Turn!

1. Choose a method of division that makes the most sense to you and find each quotient. Show your work and models.

    a. $1{,}276 \div 25$

    b. $7{,}839 \div 92$

2. The Publix Warehouse was distributing a shipment of bottled water to 36 Publix grocery stores. There was a total of $18{,}396$ packages to distribute. If each store received an equal amount of bottled water, how many packages did they receive?

1.7.6 Wrap-Up: Growth Mindset

1. Remember the brain is flexible, so your math skills can always grow and develop. Write a positive statement about what you learned in the lesson.

2. Complete the statements.

   - I do not know how to do ______________________

     _______________________________________ YET!

   - I plan to do ________________________________

     to help my math skills grow and develop.